

KILLARA HIGH SCHOOL



YEAR 12 – 2009

HSC ASSESSMENT 1

MATHEMATICS EXTENSION 2

Time allowed: 70 minutes

Total marks: 40

- NOT ALL questions are of equal value.
- Start each question on a new page.
- Show all necessary working.

Start each question on a new page

1. Given that $z = 2 + 3i$ and $w = 3 - i$, find in the form $a + ib$:

a) $z + w$ 1

b) $z \cdot \overline{w}$ 2

c) $\frac{z}{w}$ 2

2. a) Find the square root of $5 - 12i$. 2

- b) Hence, or otherwise solve the equation in the form $a + ib$ 3

$$z^2 - (1 - 4i)z - (5 - i) = 0.$$

3. By expressing z in modulus-argument form simplify

$$(2 - 2i)^{10} \cdot (\sqrt{3} + i)^6$$
 3

4. a) Find the five fifth roots of $1 + \sqrt{3}i$ 4

- b) Graph these roots on an Argand diagram and find the area of the pentagon. 2

5. If w is a complex cube root of unity, prove that

a) $1 + w + w^2 = 0$ 2

b) $\frac{a + bw + cw^2}{c + aw + bw^2} = w^2$ 2

Start each question on a new page

6. a) Draw a neat labelled sketch to indicate the locus of the complex number z

i) If $|z + 2 - i| = |z - 4|$ 3

ii) If both $\arg\left(\frac{z-4}{z}\right) > \frac{\pi}{2}$ and $0 \leq \text{im}(z) \leq 1$ are satisfied. 3

7. Complex numbers $z_1 = \frac{p}{1+2i}$ and $z_2 = \frac{q}{1+i}$ where p and q are real. 3

Find p and q If $z_1 - z_2 = 4i$

8. α and β are two complex numbers. Given that $\alpha - \beta = i(\alpha + \beta)$, on an Argand diagram

- a) neatly label and sketch:

i) α 2

ii) β

iii) $\alpha - \beta$

iv) $\alpha + \beta$

b) Find $\arg\left(\frac{\alpha + \beta}{\beta}\right)$ 2

c) If $\beta = r(\cos \theta + i \sin \theta)$, find the complex number which represents $\alpha + \beta$. 2

d) If $\theta = \frac{\pi}{6}$ and $r = 1$ in part (c). The point Q represents $\alpha + \beta$ is rotated through $\frac{11\pi}{12}$ 1

clockwise direction to become Q', Find the complex number represented by Q' in the form $a + ib$.

END OF PAPER



Q1 $z+w = 2+3+3i-i$
 $= 5+2i \checkmark$

b) $z \cdot \bar{w} = (2+3i)(3+i) \checkmark$
 $= 6+2i+9i-3$
 $= 3+11i \checkmark$

c) $\frac{z}{w} = \frac{2+3i}{3-i}$
 $= \frac{(2+3i)(3+i)}{9+1} \checkmark$
 $= \frac{3}{10} + \frac{11i}{10} \checkmark$

Q2 a) $5-12i = 9-12i+4i^2 \checkmark$
 $= (2i-3)^2 \checkmark$
 $\therefore \sqrt{5-12i} = \pm(2i-3)$

b)

$$z = \frac{1-4i \pm \sqrt{(1-4i)^2 + 4(5-i)}}{2} \checkmark$$

$$= \frac{1-4i \pm \sqrt{5-12i}}{2}$$

$$= \frac{1-4i+2i-3}{2}, \frac{1-4i-2i+3}{2}$$

$\therefore$ The 2 roots are $-(1-i) \checkmark$ and $2-3i \checkmark$

Q3 $(2-2i)^{10} = \left(\sqrt{2} \operatorname{cis} -\frac{\pi}{4} \right)^{10} \quad (1)$
 $= 2^5 \operatorname{cis} \frac{\pi}{2} = 8^5 i \checkmark$

$(\sqrt{3}+i)^6 = \left(2 \operatorname{cis} \frac{\pi}{6} \right)^6 \quad (2)$
 $= 2^6 \operatorname{cis} \pi$
 $= -2^6 \checkmark$

$(1) \times (2) = 8^5 i \times -2^6$
 $= -2097152 i \checkmark$

Q4

a) $1+\sqrt{3}i = 2 \operatorname{cis} \frac{\pi}{3}$
 $= r^5 \operatorname{cis} 5\theta$

$\therefore$ The five fifth roots of $1+\sqrt{3}i$ are of the form $2^{1/5} \operatorname{cis} \left(\frac{\pi}{15} + \frac{2k\pi}{5} \right) \checkmark$
 where $k=0,1,2,3,4 \dots$

$\therefore z_1 = 2^{1/5} \operatorname{cis} \frac{\pi}{15} \checkmark$

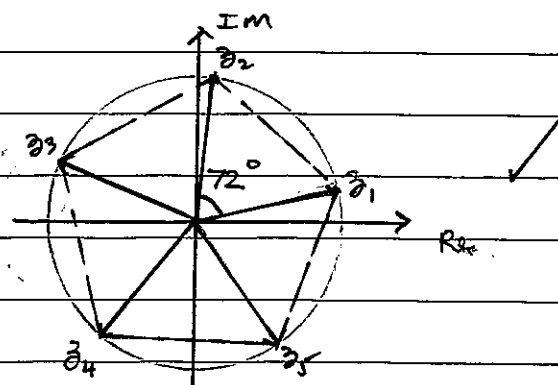
$z_2 = 2^{1/5} \operatorname{cis} \frac{7\pi}{15} \checkmark$

$z_3 = 2^{1/5} \operatorname{cis} \frac{13\pi}{15} \checkmark$

$z_5 = 2^{1/5} \operatorname{cis} -\frac{\pi}{3} \checkmark$

$z_4 = 2^{1/5} \operatorname{cis} -\frac{11\pi}{15} \checkmark$

(3)



The roots are equally spaced around the circle with $r = 2^{1/5}$, at an angle $\frac{2\pi}{5}$

b) Area of pentagon
 $A = 5 \left(\frac{1}{2} \times 2^{1/5} \times 2^{1/5} \sin \frac{2\pi}{5} \right)$

$= 3.14 \text{ sq unit.} \checkmark$



Q5

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(a)

The 3 cube roots of unity are

$$-\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad -\frac{1}{2} - \frac{\sqrt{3}}{2}i \quad \text{and} \quad 1$$

$$\therefore 1 + \omega + \omega^2 = 1 - \frac{1}{2} + \frac{\sqrt{3}}{2}i - \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$$= 0$$

$$= \text{R.H.S} \quad \checkmark$$

Alternatively

The equation for cube roots of unity is

$$z^3 - 1 = 0$$

$$(z-1)(z^2+z+1) = 0 \quad \checkmark$$

$$\omega \neq 1 \therefore z-1 \neq 0$$

$$\text{hence } z^2+z+1 = 0 \quad \checkmark$$

$$\therefore \omega^2 + \omega + 1 = 0$$

(b)

$$\text{L.H.S} = \frac{(a+b\omega+c\omega^2) \times \omega^3}{(c+a\omega+b\omega^2) \times \omega^3} \quad \checkmark$$

$$= \frac{\omega^2(a\omega+b\omega^2+c\omega^3)}{c\omega^3+a\omega^4+b\omega^5}$$

$$\text{but } \omega^3=1, \omega^4=\omega \text{ and } \omega^5=\omega^2$$

$$\therefore \text{L.H.S} = \frac{\omega^2(a\omega+b\omega^2+c)}{c+a\omega+b\omega^2} \quad \checkmark$$

$$= \omega^2 = \text{R.H.S}$$

Q6

$$\text{a) } |z+2-i| = |z-4|$$

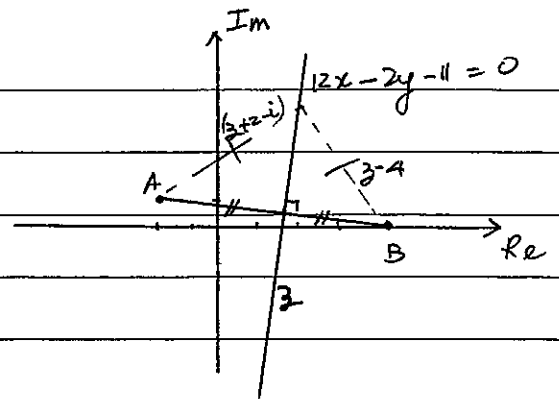
$$(x+2)^2 + (y-1)^2 = (x-4)^2 + y^2$$

$$x^2+4x+4+y^2-2y+1 = x^2-8x+16+y^2$$

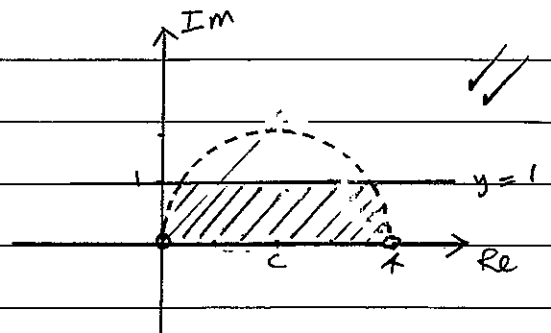
$$12x-2y-11=0$$

$\therefore$ locus of z is the
bisector to AB with the
equation $12x-2y-11=0$

✓✓



(b)



$\text{Arg}\left(\frac{z-4}{z}\right) = \frac{\pi}{2}$ is the semicircle with radius $r=2$, centre $C(2,4)$.

$\text{Arg}\left(\frac{z-4}{z}\right) > \frac{\pi}{2}$ is inside the semi-circle.

Q7

$$z_1 - z_2 = \frac{p}{1+2i} - \frac{q}{1+i}$$

$$= \frac{p(1-2i)}{5} - \frac{q(1-i)}{2}$$

$$= \frac{2p-4pi-5q-5qi}{10} = 4i$$

$$\therefore \frac{2p-5q}{10} = 0 \quad \text{and} \quad \frac{4p-5q}{10} = -4$$

$$\Rightarrow 4p-5q = 40$$

$$2p-5q = 0$$

$$2p = -40 \Rightarrow p = -20 \quad \checkmark$$

and

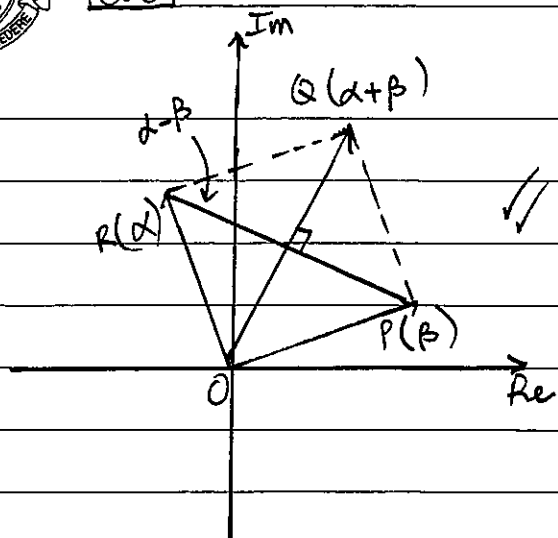
$$-40-5q = 0 \Rightarrow q = -8 \quad \checkmark$$



Q8

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(a)



(b)

Since $\alpha - \beta = i(\alpha + \beta)$, from the diagram

$\vec{PR}$ is the rotation of $\vec{OQ}$, anticlockwise, by $\frac{\pi}{2}$, about O.

$\therefore PR = OQ$

$\therefore OPQR$ is a square (diags' are equal & bisect at $\perp$)

hence $\angle QOP = \frac{\pi}{4}$ ✓

$$\arg\left(\frac{\alpha+\beta}{\beta}\right) = \arg(\alpha+\beta) - \arg\beta$$

$$= \angle QOP$$

$$= \frac{\pi}{4} \quad \checkmark$$

(diag. of a square bisects the angle which it passes.)

(c)

$$|\alpha+\beta| = \sqrt{r^2+r^2}$$

$$= r/\sqrt{2} \quad \checkmark$$

$$\arg(\alpha+\beta) = \arg\beta + \angle QOP$$

$$= \theta + \frac{\pi}{4}$$

$$\therefore \alpha+\beta = r/\sqrt{2} \operatorname{cis}\left(\theta + \frac{\pi}{4}\right) \quad \checkmark$$

(d) if $\theta = \frac{\pi}{6}$, $r = 1$

$$\text{then } \alpha+\beta = \sqrt{2} \operatorname{cis} \frac{5\pi}{12}$$

$\therefore$ The complex number represents Q' is

$$\sqrt{2} \operatorname{cis}\left(\frac{5\pi}{12} - \frac{11\pi}{12}\right) = \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{2}\right)$$

$$= \sqrt{2} \left(\cos -\frac{\pi}{2} + i \sin -\frac{\pi}{2}\right)$$

$$= -\sqrt{2} i \quad \checkmark$$